Fiscal Year:	FY 2007	**Task Last Updated:**	FY 12/31/2006
PI Name:	Sundaresan, Alamelu Ph.D.		
Project Title:	Targeting specific bone cell signaling pathways and immunosuppression in microgravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77004	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	UNSOL-JSC
Start Date:	01/01/2006	**End Date:**	12/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Pellis, Neal	**Contact Phone:**	
Contact Email:	neal.r.pellis@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ06HA40G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The utility of the above project is apparent, in view of the high level of importance placed by NASA on understanding and developing countermeasures to the effects of microgravity exposure and consequent unloading-induced bone loss and immune suppression in astronauts. Both these projects could provide novel information about the effects of unloading on the complex cellular interactions between cells of the skeletal system and the potential use of pharmacological countermeasures such as statins and bisphosphonates. The signaling experiments would delineate target molecules which could be manipulated to maintain equilibrium even in unloaded conditions. Cellular and tissue level changes occur much earlier than whole organ changes, which can be addressed in both these projects, thereby providing a mechanistic understanding (cell signaling modulation within the bone matrix/immune cell complex) and basis for developing whole body countermeasures for alleviating bone loss and immune suppression from the space flight		

	perspective. Both these projects will involve travel to national and international NASA related and basic science related symposia by the PI.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Once the Wnt signaling pathways are completed, the significance of this biomarker for osteoporosis and Lupus etiologies can be investigated.
Task Progress:	The project was slated to begin in January and is funded from Jan-Dec 2006. It is in its tenth month. The PI began to work on protocol optimization and experimental set up. The protocols for immunofluorescence and cell culture have been completed. The request for the antibody from University of Southwestern Wales has been sent with appropriate letters to the US dept of customs, regarding antibody storage. One graduate student has been recruited for the project with recruitment of two summer interns and one other graduate student from Texas Southern University. The summer interns will start at the end of May 2006. Two sets of experiments with normal human lymphocytes were completed in March 2006. Total RNA is being isolated from these samples. Primers for key upstream lymphocyte targets were prepared by the PI after extensive gene bank data base searches and primer preparation strategies. The primers were then prepared using a DNA synthesizer. After completion of some protein experiments which were promising and reproducible, an abstract was submitted to the Society of in vitro biology and accepted for a presentation in June, 2006. This summer entailed detailed training of the students in laboratory techniques and beginning of proposed real time PCR experiments. There have been two major hurdles in the project. The PI has been unsuccessful in acquiring regular access to JSC facilities where the project is to be based and has had to complete some experiments at TSU and the UT Medical School. The other major issue has been revamping of facilities at JSC and lack of incubators and hoods for tissue culture until March 2006. The summer tasks allocated resulted in the confirmation of previous gene array experiments with real time PCR at TSU. The results were promising and presented at the NASA URC presentation. Dr. Sundaresan mentored two students from the NASA URC intern program, Jamail Plumber and Elise Nguyen. Both undergraduate students had a fulfilling summer experience and are continuing to work with Dr.Sundaresan. One manuscript is in press in the Journal of “Microgravity Science and Technology.” Bone cell experiments will commence once the laboratory has been set up for tissue culture (Dec 06). The signaling targets and bone cell growth with different natural and pro bone formation products will be assessed “in vitro.” Experiments with the “Wnt” gene and Serotonin are in progress at NASA and UT Medical School.
Bibliography Type:	Description: (Last Updated: 07/02/2026)
Abstracts for Journals and Proceedings	Sundaresan A (Lalita) , Singh K , Pellis NR, DuMond J Jr. "Signal transduction targets of modeled microgravity." 2006 In Vitro Biology Meeting, Minneapolis, MN, June 3-7, 2006. In Vitro Biology Meeting, Abstracts, p. 27A, 2006. , Jun-2006
Articles in Peer-reviewed Journals	Sundaresan A, Kulkarni AD, Yamauchi K and Pellis NR. "THE ROLE OF NUCLEOTIDES IN AUGMENTATION OF LYMPHOCYTE LOCOMOTION: ADAPTATIONAL COUNTERMEASURE DEVELOPMENT IN MICROGRAVITY ANALOG ENVIRONMENTS: " Microgravity Science and Technology, in press, December 2006. , Dec-2006